C Programming

Kai Du

Lecture 2

Lecture 2: Outline

- **Variables, Data Types, and Arithmetic Expressions [K- ch.4]**
  - Working with Variables
  - Understanding Data Types and Constants
    - The Basic Integer Type int
    - The Floating Number Type float
    - The Extended Precision Type double
    - The Single Character Type char
    - The Boolean Data Type _Bool
    - Storage sizes and ranges
    - Type Specifiers: long, long long, short, unsigned, and signed
  - Working with Arithmetic Expressions
    - Integer Arithmetic and the Unary Minus Operator
    - The Modulus Operator
    - Integer and Floating-Point Conversions
  - Combining Operations with Assignment: The Assignment Operators
  - Types _Complex and _Imaginary

Variables

- Programs can use symbolic names for storing computation data
- **Variable: a symbolic name for a memory location**
  - programmer doesn't have to worry about specifying (or even knowing) the value of the location's address
- In C, variables have to be ***declared*** before they are used
  - Variable declaration: [symbolic name(*identifier*), type]
- Declarations that reserve storage are called ***definitions***
  - The definition reserves memory space for the variable, but doesn't put any value there
- Values get into the memory location of the variable by ***initialization*** or ***assignement***

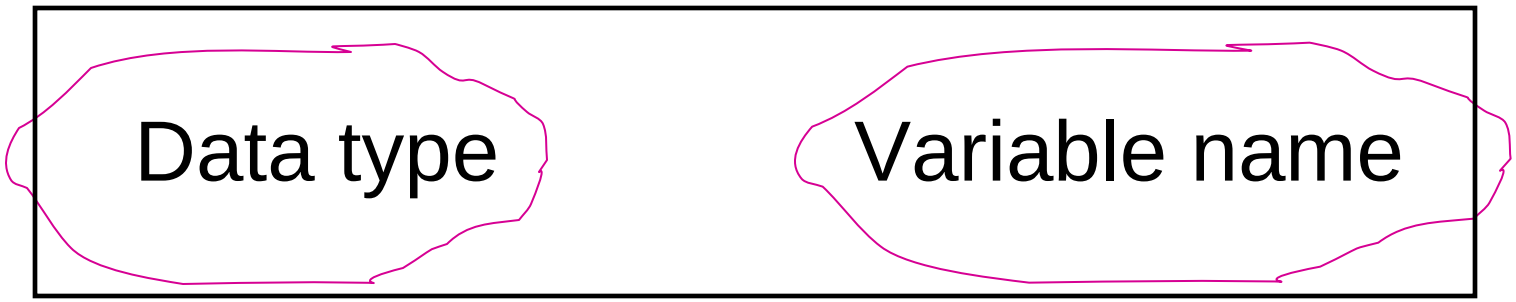
Variables - Examples

```
int a;          // declaring a variable of type int

int sum, a1,a2; // declaring 3 variables

int x=7; // declaring and initializing a variable


a=5;  // assigning to variable a the value 5

a1=a; // assigning to variable a1 the value of a
```

L-value ***R-value***

```
a1=a1+1;  // assigning to variable a1 the value of a1+1
          // (increasing value of a1 with 1)
```

Variable declarations



Which data types are possible in C ?

Which variable names are allowed in C ?

Variable names

Rules for valid variable names (*identifiers*) in C :

- Name must begin with a letter or underscore (_) and can be followed by any combination of letters, underscores, or digits.
- Any name that has special significance to the C compiler (*reserved words*) cannot be used as a variable name.
- Examples of ***valid*** variable names: `Sum`, `pieceFlag`, `I`, `J5x7`, `Number_of_moves`, `_sysflag`
- Examples of ***invalid*** variable names: `sum$value`, `3Spencer`, `int`.
- C is case-sensitive: `sum`, `Sum`, and `SUM` each refer to a different variable !
- Variable names can be as long as you want, although only the first 63 (or 31) characters might be significant. (Anyway, it's not practical to use variable names that are too long)
- Choice of meaningful variable names can increase the readability of a program

Data types

- **Basic data types in C: `int`, `float`, `double`, `char`, and `_Bool`.**
- Data type `int:` can be used to store integer numbers (values with no decimal places)
- Data type type `float:` can be used for storing floating-point numbers (values containing decimal places).
- Data type `double:` the same as type `float`, only with roughly twice the precision.
- Data type `char:` can be used to store a single character, such as the letter *a*, the digit character *6*, or a semicolon.
- Data type `_Bool:` can be used to store just the values 0 or 1 (used for indicating a true/false situation). This type has been added by the C99 standard (was not in ANSI C)

Example: Using data types

```
#include <stdio.h>
int main (void)
{
    int integerVar = 100;
    float floatingVar = 331.79;
    double doubleVar = 8.44e+11;
    char charVar = 'W';
    _Bool boolVar = 0;
    printf ("integerVar = %i\n", integerVar);
    printf ("floatingVar = %f\n", floatingVar);
    printf ("doubleVar = %e\n", doubleVar);
    printf ("doubleVar = %g\n", doubleVar);
    printf ("charVar = %c\n", charVar);
    printf ("boolVar = %i\n", boolVar);
    return 0;
}
```

The basic data type `int`

- Examples of integer constants: 158, –10, and 0
- No embedded spaces are permitted between the digits, and values larger than 999 cannot be expressed using commas. (The value 12,000 is not a valid integer constant and must be written as 12000.)
- Integer values can be displayed by using the format characters `%i` in the format string of a `printf` call.
- Also the `%d` format characters can be used to display an integer (Kernighan&Ritchie C)
- **Integers can also be expressed in a base other than decimal (base 10): octal (base 8) or hexa (base 16).**

Octal notation for integers

- **Octal notation** (base 8): If the first digit of the integer value is 0, the integer is taken as expressed in *octal* notation. In that case, the remaining digits of the value must be valid base-8 digits and, therefore, must be 0–7.
- Example: Octal value `0177` represents the decimal value `127` ($1 \times 8^2 + 7 \times 8 + 7$).
- An integer value can be displayed in octal notation by using the format characters `%o` or `%#o` in the format string of a `printf` statement.

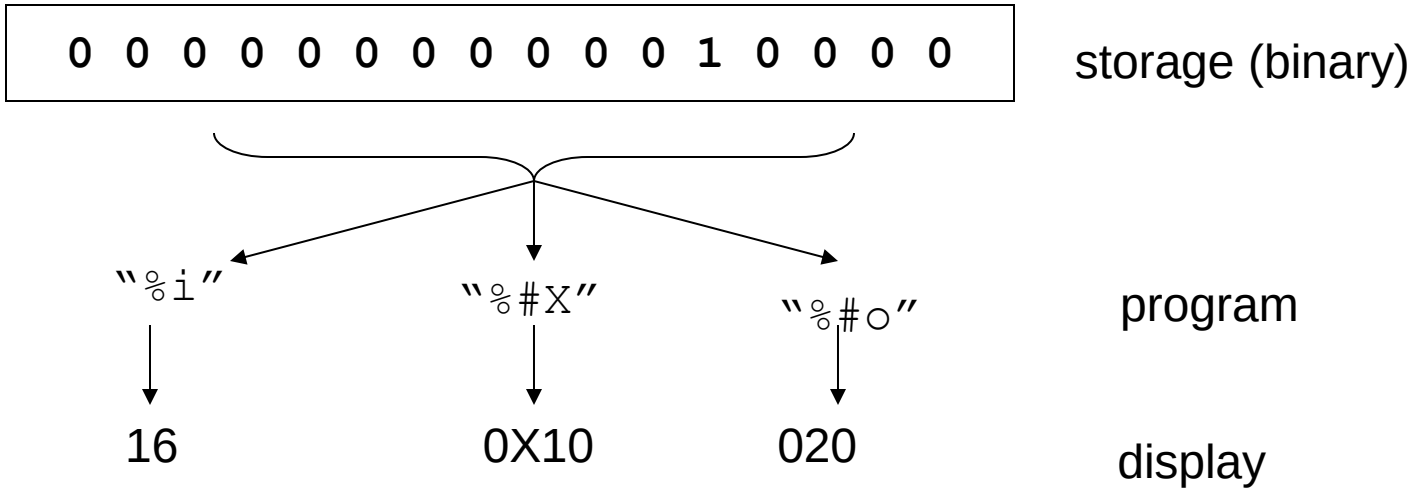
Hexadecimal notation for integers

- **Hexadecimal notation** (base 16): If an integer constant is preceded by a zero and the letter *x* (either lowercase or uppercase), the value is taken as being expressed in hexadecimal. Immediately following the letter *x* are the digits of the hexadecimal value, which can be composed of the digits 0–9 and the letters a–f (or A–F). The letters represent the values 10–15, respectively.
- Example: hexadecimal value `0xA3F` represents the decimal value `2623` ($10 \times 16^2 + 3 \times 16 + 15$).
- The format characters `%x`, `%X` , `%#x`, or `%#X` display a value in hexadecimal format

Data display vs data storage

- **The option to use decimal, octal or hexadecimal notation doesn't affect how the number is actually stored internally !**
- When/where to use octal and hexa: to express computer-related values in a more convenient way

```
int x =16;
printf("%i %#X %#o\n", x,x,x);
```



The floating number type `float`

- A variable declared to be of type `float` can be used for storing values containing decimal places.
- Examples of floating-point constants : 3., 125.8, –.0001
- To display a floating-point value at the terminal, the `printf` conversion characters `%f` are used.
- Floating-point constants can also be expressed in *scientific notation*.The value `1.7e4` represents the value 1.7×10^4.
- The value before the letter e is known as the *mantissa*, whereas the value that follows is called the *exponent*. This exponent, which can be preceded by an optional plus or minus sign, represents the power of 10 by which the mantissa is to be multiplied.
- To display a value in scientific notation, the format characters `%e` should be specified in the `printf` format string.
- The `printf` format characters `%g` can be used to let `printf` decide whether to display the floating-point value in normal floating-point notation or in scientific notation.This decision is based on the value of the exponent: If it's less than –4 or greater than 5, `%e` (scientific notation) format is used; otherwise, `%f` format is used.

The extended precision type `double`

- Type `double` is very similar to type `float`, but it is used whenever the range provided by a `float` variable is not sufficient. Variables declared to be of type `double` can store roughly twice as many significant digits as can a variable of type `float`.
- Most computers represent double values using 64 bits.
- Unless told otherwise, all floating-point constants are taken as `double` values by the C compiler!
- To explicitly express a `float` constant, append either an f or F to the end of the number: `12.5f`
- To display a `double` value, the format characters `%f`, `%e`, or `%g`, which are the same format characters used to display a `float` value, can be used.

The character type `char`

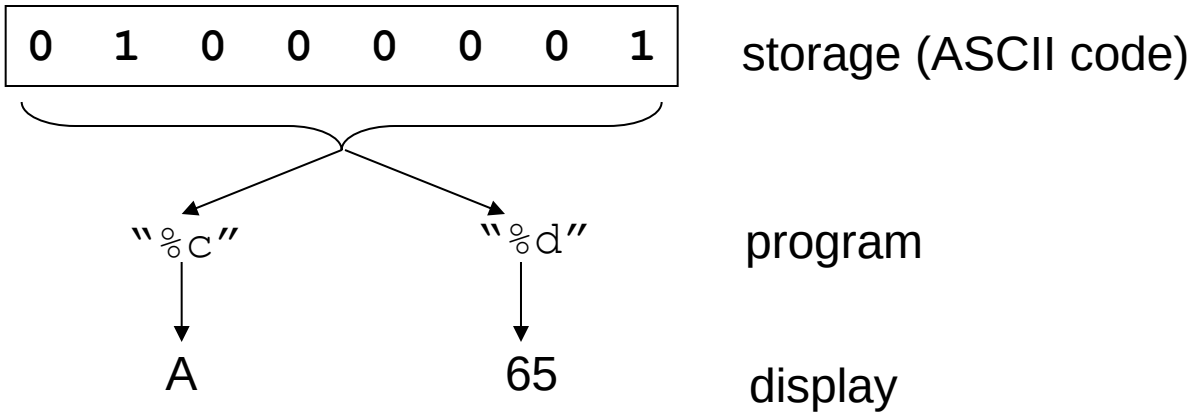
- A `char` variable can be used to store a single character.
- A character constant is formed by enclosing the character within a pair of single quotation marks. Valid examples: 'a', ';', and '0‘.
- Character zero (`'0'`) is not the same as the number (integer constant) `0`.
- Do not confuse a character constant with a character string: character `'0'` and string `"0"`.
- The character constant ‘`\n`’—the newline character—is a valid character constant : the backslash character is a special character in the C system and does not actually count as a character.
- There are other special characters *(escape sequences*) that are initiated with the backslash character: \\, \”, \t
- The format characters `%c` can be used in a `printf` call to display the value of a `char` variable
- To handle characters internally, the computer uses a numerical code in which certain integers represent certain characters. The most commonly used code is the ASCII code

Assigning values to char

```
char letter;  /* declare  variable letter of type char */

letter = 'A';    /* OK */
letter = A;      /* NO! Compiler thinks A is a variable */
letter = "A";    /* NO! Compiler thinks "A" is a string */
letter = 65;     /* ok because characters are really
                    stored as numeric values (ASCII code),
                    but poor style */
```

Data display vs data storage

```
/* displays ASCII code for a character */

#include <stdio.h>
int main(void)
{
    char ch;
    ch='A';
    printf("The code for %c is %i.\n", ch, ch);
    return 0;
}
```



The Boolean Data Type `_Bool`

- A `_Bool` variable is defined in the language to be large enough to store just the values 0 and 1.The precise amount of memory that is used is unspecified.
- `_Bool` variables are used in programs that need to indicate a Boolean condition. For example, a variable of this type might be used to indicate whether all data has been read from a file.
- By convention, 0 is used to indicate a false value, and 1 indicates a true value. When assigning a value to a `_Bool` variable, a value of 0 is stored as 0 inside the variable, whereas any nonzero value is stored as 1.
- To make it easier to work with `_Bool` variables in your program, the standard header file `<stdbool.h>` defines the values `bool`, `true`, and `false`:

```
bool endOfData = false;
```

- The `_Bool` type has beed added by C99.
- Some compilers (Borland C, Turbo C, Visual C) don’t support it

Storage sizes and ranges

- **Every type has a *range* of values associated with it**.
- This range is determined by the amount of storage that is allocated to store a value belonging to that type of data.
- In general, that amount of storage is not defined in the language. It typically depends on the computer you're running, and is, therefore, called *implementation-* or *machine*-dependent.
  - For example, an integer might take up 32 bits on your computer, or it might be stored in 64.You should never write programs that make any assumptions about the size of your data types !
- The language standards only guarantees that a minimum amount of storage will be set aside for each basic data type.
  - For example, it's guaranteed that an integer value will be stored in a minimum of 32 bits of storage, which is the size of a "word" on many computers.

Integer overflow

- What happens if an integer tries to get a value too big for its type (out of range)?

```
#include <stdio.h>
int main(void) {
    int i = 2147483647;
    printf("%i %i %i\n", i, i+1, i+2);
    return 0;
}
```

Program output:

```
2147483647 -2147483648 -2147483647
```

Explanation:
On this computer, int is stored on 32 bits: the first bit represents the sign, the rest of 31 bits represent the value.
Biggest positive int value here: $2^{31}-1 = 2147483647$

Floating point round-off error

```
#include <stdio.h>
int main(void)
{
    float a,b;
    b = 2.0e20 + 1.0;
    a = b - 2.0e20;
    printf("%f \n", a);
    return 0;
}
```

Program output:

```
4008175468544.000000
```

Explanation: the computer doesn't keep track of enough decimal places ! The number 2.0e20 is 2 followed by 20 zeros and by adding 1 you are trying to change the 21st digit. To do this correctly, the program would need to be able to store a 21-digit number. A float number is typically just six or seven digits scaled to bigger or smaller numbers with an exponent.

Type Specifiers: `long`, `long long`, `short`, `unsigned`, `signed`

- Type specifiers: extend or limit the range of certain basic types on certain computer systems
- If the specifier `long` is placed directly before the `int` declaration, the declared integer variable is of extended range on some computer systems.
- Example of a `long int` declaration: `long int factorial;`
- On many systems, an `int` and a `long int` both have the same range and either can be used to store integer values up to 32-bits wide (2^{31}–1, or 2,147,483,647).
- A constant value of type `long int` is formed by optionally appending the letter L (upper- or lowercase) at the end of an integer constant.
- Example: `long int numberOfPoints = 131071100L;`
- To display the value of a `long int` using `printf`, the letter l is used as a modifier before the integer format characters i, o, and x

Basic Data Types - Summary

Type	Meaning	Constants Ex.	printf
int	Integer value; guaranteed to contain **at least 16 bits**	`12, -7, 0xFFE0, 0177`	`%i,%d, %x, %o`
short int	Integer value of reduced precision; guaranteed to contain **at least 16 bits**	-	`%hi, %hx, %ho`
long int	Integer value of extended precision; guaranteed to contain **at least 32 bits**	`12L, 23l, 0xffffL`	`%li, %lx, %lo`
long long int	Integer value of extraextended precision; guaranteed to contain **at least 64 bits**	`12LL, 23ll, 0xffffLL`	`%lli, %llx, %llo`
unsigned int	Positive integer value; can store positive values up to twice as large as an int; guaranteed to contain at least 16 bits (all bits represent the value, no sign bit)	`12u, 0XFFu`	`%u, %x, %o`
unsigned short int		-	`%hu, %hx, %ho`
unsigned long int		`12UL, 100ul, 0xffeeUL`	`%lu, %lx, %lo`
unsigned long long int		`12ull, 0xffeeULL`	`%llu, %llx, %llo`

Basic Data Types - Summary (contd.)

Type	Meaning	Constants	printf
float	Floating-point value; a value that can contain decimal places; guaranteed to contain at **least six digits of precision**.	`12.34f, 3.1e-5f`	`%f, %e, %g`
double	Extended accuracy floating-point value; guaranteed to contain **at least 10 digits of precision**.	`12.34, 3.1e-5,`	`%f, %e, %g`
long double	Extraextended accuracy floating-point value; guaranteed to contain **at least 10 digits of precision.**	`12.341, 3.1e-5l`	`%Lf, %Le, %Lg`
char	Single character value; on some systems, sign extension might occur when used in an expression.	`'a', '\n'`	`%c`
unsigned char	Same as char, except ensures that sign extension does not occur as a result of integral promotion.	-	
signed char	Same as char, except ensures that sign extension does occur as a result of integral promotion.	-	

Knowing actual ranges for types

- Defined in the system include files `<limits.h>` and `<float.h>`
- `<limits.h>` contains system-dependent values that specify the sizes of various character and integer data types:
  - the maximum size of an `int` is defined by the name `INT_MAX`
  - the maximum size of an `unsigned long int` is defined by `ULONG_MAX`
- `<float.h>` gives information about floating-point data types.
  - `FLT_MAX` specifies the maximum floating-point number,
  - `FLT_DIG` specifies the number of decimal digits of precision for a float type.

Working with arithmetic expressions

- Basic arithmetic operators: +, -, *, /
- **Precedence**: one operator can have a higher priority, or *precedence*, over another operator.
  - Example: * has a higher precedence than +
  - a + b * c
  - if necessary, you can always use parentheses in an expression to force the terms to be evaluated in any desired order.
- **Associativity**: Expressions containing operators of the same precedence are evaluated either from left to right or from right to left, depending on the operator. This is known as the *associative* property of an operator
  - Example: + has a *left to right* associativity

- In C there are many more operators -> later in this course !
- (Table A5 in Annex A of [Kochan]: full list, with precedence and associativity for all C operators)

Working with arithmetic expressions

```
#include <stdio.h>
int main (void)
{
    int a = 100;
    int b = 2;
    int c = 25;
    int d = 4;
    int result;
    result = a - b; // subtraction
    printf ("a - b = %i\n", result);
    result = b * c; // multiplication
    printf ("b * c = %i\n", result);
    result = a / c; // division
    printf ("a / c = %i\n", result);
    result = a + b * c; // precedence
    printf ("a + b * c = %i\n", result);
    printf ("a * b + c * d = %i\n", a * b + c * d);
    return 0;
}
```

Integer arithmetic and the unary minus operator

```
// More arithmetic expressions
#include <stdio.h>
int main (void)
{
    int a = 25;
    int b = 2;
    float c = 25.0;
    float d = 2.0;
    printf ("6 + a / 5 * b = %i\n", 6 + a / 5 * b);
    printf ("a / b * b = %i\n", a / b * b);
    printf ("c / d * d = %f\n", c / d * d);
    printf ("-a = %i\n", -a);
    return 0;
}
```

The modulus operator

```
// The modulus operator
#include <stdio.h>
int main (void)
{
    int a = 25, b = 5, c = 10, d = 7;
    printf ("a %% b = %i\n", a % b);
    printf ("a %% c = %i\n", a % c);
    printf ("a %% d = %i\n", a % d);
    printf ("a / d * d + a %% d = %i\n",
                                    a / d * d + a % d);
    return 0;
}
```

Modulus operator: %
Binary operator
Gets the remainder resulting from integer division
% has equal precedence to * and /

Integer and Floating-Point Conversions

- Assign an integer value to a floating variable: does not cause any change in the value of the number; the value is simply converted by the system and stored in the floating
- Assign a floating-point value to an integer variable: the decimal portion of the number gets truncated.
- Integer arithmetic (division):
  - int divided to int => result is integer division
  - int divided to float or float divided to int => result is real division (floating-point)

Integer and Floating-Point Conversions

```
// Basic conversions in C
#include <stdio.h>
int main (void)
{
float f1 = 123.125, f2;
int i1, i2 = -150;
char c = 'a';
i1 = f1; // floating to integer conversion
printf ("%f assigned to an int produces %i\n", f1, i1);
f1 = i2; // integer to floating conversion
printf ("%i assigned to a float produces %f\n", i2, f1);
f1 = i2 / 100; // integer divided by integer
printf ("%i divided by 100 produces %f\n", i2, f1);
f2 = i2 / 100.0; // integer divided by a float
printf ("%i divided by 100.0 produces %f\n", i2, f2);
f2 = (float) i2 / 100; // type cast operator
printf ("(float) %i divided by 100 produces %f\n", i2, f2);
return 0;
}
```

The Type Cast Operator

- `f2 = (float) i2 / 100; // type cast operator`
- The type cast operator has the effect of converting the value of the variable i2 to type float for purposes of evaluation of the expression.
- This operator does NOT permanently affect the value of the variable i2;
- **The type cast operator has a higher precedence than all the arithmetic operators except the unary minus and unary plus**.

- Examples of the use of the type cast operator:
- `(int) 29.55 + (int) 21.99` results in `29 + 21`
- `(float) 6 / (float) 4` results in `1.5`
- `(float) 6 / 4` results in `1.5`

The assignment operators

- The C language permits you to join the arithmetic operators with the assignment operator using the following general format: `op=`, where `op` is an arithmetic operator, including +, –, ×, /, and %.
- `op` can also be a logical or bit operator => later in this course
- Example:

```
count += 10;
```

  - Equivalent with:

```
count=count+10;
```

- Example: precedence of op=:

```
a /= b + c
```

  - Equivalent with:

```
a = a / (b + c)
```

  - addition is performed first because the addition operator has higher precedence than the assignment operator

_Complex and _Imaginary types

- Supported only by a few compilers
- Complex numbers have two components: a real part and an imaginary part. C99 represents a complex number internally with a two-element array, with the first component being the real part and the second component being the imaginary part.

- There are three complex types:
- `float _Complex` represents the real and imaginary parts with type float values.
- `double _Complex` represents the real and imaginary parts with type double values.
- `long double _Complex` represents the real and imaginary parts with type long double values.

- There are three imaginary types; An imaginary number has just an imaginary part:
- `float _Imaginary` represents the imaginary part with a type float value.
- `double _Imaginary` represents the imaginary part with a type double value.
- `long double _Imaginary` represents the imaginary part with a type long double value.

- Complex numbers can be initialized using real numbers and the value `I`, defined in `<complex.h>` and representing i, the square root of –1

_Complex example

```
#include <complex.h> // for I
int main(void) {
    double _Complex z1 = 3*I;
    double _Complex z2=5+4*I;
    double _Complex z;
    z=z1*z2;
    printf("%f+%f*I\n",creal(z),cimag(z) );
}
```

Declaring variables

- Some older languages (FORTRAN, BASIC) allow you to use variables without declaring them.
- Other languages (C, Pascal) impose to declare variables
- **Advantages of languages with variable declarations:**
  - Putting all the variables in one place makes it easier for a reader to understand the program
  - Thinking about which variables to declare encourages the programmer to do some planning before writing a program (What information does the program need? What must the program to produce as output? What is the best way to represent the data?)
  - The obligation to declare all variables helps prevent bugs of misspelled variable names.
  - **Compiler knows the amount of statically allocated memory needed**
  - **Compiler can verify that operations done on a variable are allowed by its type (*strongly typed languages*)**

Declaration vs Definition

- Variable declaration: [Type, Identifier]
- Variable definition: a declaration which does also reserve storage space (memory) !
  - Not all declarations are definitions
  - In the examples seen so far, all declarations are as well definitions
  - Declarations which are not definitions: later in this semester !